

Goserelin for breast cancer (Zoladex®, Zoladex LA®)

Goserelin is a cancer drug. It is used to treat breast cancer.

On this page

- [What is goserelin \(Zoladex®, Zoladex LA®\)?](#)
- [How goserelin works](#)
- [Having goserelin](#)
- [About side effects](#)
- [Very common side effects](#)
- [Other side effects](#)
- [Other important information](#)
- [How we can help](#)

What is goserelin (Zoladex®, Zoladex LA®)?

Goserelin is a hormonal therapy drug. It is also called Zoladex® or Zoladex LA®. It is used to treat [breast cancer](#).

It is best to read this information with our general information about [hormonal therapy](#) and the type of cancer you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

This information is for anyone having goserelin as part of treatment for breast cancer. Goserelin can also be used to treat prostate cancer. We have separate information about [goserelin for prostate cancer](#).

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How goserelin works

Hormones are chemicals that our bodies make. They act as messengers and help control how cells and organs work. Hormonal therapy drugs change how hormones are made or work in the body.

Many breast cancers rely on the hormone oestrogen to grow. This type of breast cancer is called oestrogen-receptor positive (ER positive) breast cancer. Goserelin stops the body making oestrogen. This lowers oestrogen levels, which may stop breast cancer growing, or shrink the cancer.

Having goserelin

You have goserelin as a small pellet (implant) injected under the skin (subcutaneously) of your tummy (abdomen). The pellet releases the drug slowly as it dissolves under the skin.

During your treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information. During your course of treatment, you may also have blood tests.

Zoladex® is given every 4 weeks. Zoladex LA® is given every 12 weeks.

You usually have the first injection at your hospital clinic appointment. After this your practice nurse or GP can give it to you at the surgery. They may use a local anaesthetic cream to numb the area before the injection.

If you are not able to visit the surgery, a district nurse may give you the injection at home.

If you are taking any medicines to thin your blood, tell your doctor, nurse or pharmacist. These medicines can make you bruise more easily.

You may have goserelin with other cancer drugs. Your cancer team will discuss your treatment plan with you.

They may give you other medicines to take home. Take all your medicines exactly as they tell you to. Do not stop taking any of your medicines unless your doctor tells you to.

Injection site

The skin in the area that was injected may become:

- painful
- swollen
- red, if you have white skin
- darker, if you have black or brown skin.

If this happens, let your doctor know. Painkillers may help.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

When you have this treatment, you will have regular appointments with a doctor, nurse or pharmacist. Always tell them about any side effects you have. They can give you:

- drugs to help control most side effects
- advice about managing side effects.

Most side effects can be managed. But sometimes side effects are harder to control. It is important not to stop taking hormonal therapy without telling your doctor. If side effects cannot be managed, your doctor may suggest a different type of hormonal therapy.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Hot flushes and sweats

Hot flushes are a common side effect of this treatment. During a flush, your neck and face may feel warm. If you have white skin, your skin may look red. If you have black or brown skin, your skin colour may deepen. Flushes can last up to 5 minutes. You might also have sweats or feel anxious or irritable during a flush.

Here are some things you can do to reduce the effects of hot flushes:

-

Wear natural fabrics, such as cotton.

- Wear layers so you can remove clothes as needed.
- Use cotton sheets and layers of bedding you can remove.
- Try cooling pads or pillows to keep you cool.
- Keep rooms cool or use a fan.
- Have cold drinks and avoid caffeine and alcohol.

You might have fewer hot flushes as your body adjusts to this treatment. You could try therapies to help you [cope with hot flushes](#), such as:

- [talking therapies](#) like cognitive behaviour therapy (CBT)
- controlled breathing exercises
- yoga
- acupuncture.

If they do not improve, talk to your doctor. Certain drugs can help to improve hot flushes.

Flushes usually stop a few months after treatment ends. But some people continue to have them.

Vaginal dryness

This treatment can cause vaginal dryness and a lower sex drive. Non-hormonal creams, gels or lubricants can help with vaginal dryness and any discomfort during [sex](#). You can buy these from a chemist, or your doctor can prescribe them.

Erection problems

It is common to lose your [sex drive](#) during hormonal therapy. This can cause erection problems. Things may return to normal after you stop taking the drug. But some people continue to have problems after treatment is over. Your doctor can prescribe treatments to help with [erection problems](#). But these treatments will not increase your sex drive.

Breast swelling or tenderness

You may get swelling and tenderness of your breast tissue. This is called gynaecomastia. Your doctor can give you more advice.

Skin changes

This treatment can [affect your skin](#). It may cause a rash, which may be itchy. Your doctor or nurse can tell you what to expect. If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Always tell your doctor or nurse about any skin changes. They can give you advice and may prescribe creams or medicines to help. Changes to your skin usually improve when treatment ends.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Mood changes

You may feel low or depressed, or have mood swings, during this treatment. Talking to family and friends about how you feel might help. If mood changes last for more than a few weeks, tell your doctor, nurse or pharmacist. They can talk to you about different ways to manage low mood or [depression](#).

Bone thinning (osteoporosis)

Taking this treatment for several months or more can increase the risk of [bone thinning](#). This is called osteoporosis. It can make you more likely to get a broken bone (fracture).

You may have bone density scans to check your bone health before and during treatment. These scans can diagnose osteoporosis or osteopenia (low bone density) so that it can be monitored. If your scan shows osteoporosis, your doctor usually prescribes:

- drugs to protect your bones called [bisphosphonates](#)
- calcium and vitamin D supplements.

Regular exercise where you support your body weight, like walking, running and dancing, can help look after your bones. This is called [weight-bearing exercise](#). Other types of strength exercises are also helpful. If you have any bone thinning or the cancer is affecting your bones, get advice from your cancer team first.

Eating a healthy diet and not smoking also helps take care of your bones.

Vaginal bleeding and effects on periods

This treatment can cause irregular vaginal bleeding for up to a month after the first injection. After this, periods will usually stop. If you have vaginal bleeding, tell your doctor or nurse.

Even with no periods, you can still become pregnant if you have not yet been through menopause. You need to use non-hormonal contraception to prevent a pregnancy while having goserelin. Your doctor or nurse can tell you more about this.

Monthly periods may start again when you stop goserelin treatment. But if you had a natural menopause during treatment, your periods may not come back.

Hair thinning

While you are having this treatment, your hair may become thinner, drier and more brittle. This is usually mild. If you are worried about this, ask your nurse for advice on hair care.

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Blood pressure changes

This treatment may cause low or high blood pressure. If you have ever had any problems with your blood pressure, tell your doctor or nurse. Let them know if you feel dizzy or have any headaches.

Raised blood sugar levels

This treatment can raise your blood sugar levels and increase the risk of diabetes.

Signs of raised blood sugar include:

- feeling thirsty
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.

Tell your doctor or nurse if you have these symptoms. They can do tests to check your blood sugar level and give you advice.

If you already have [diabetes](#), your blood sugars may be higher than usual. You may need to check them more often, or make changes to your diabetic treatment. Your diabetic doctor or nurse will talk to you about how to manage this.

Muscle or joint pain

You may get pain in your muscles or joints. If this happens, tell your doctor or nurse. They can give you advice and painkillers or other drugs to help. Being physically active and keeping to a healthy weight may help with the pain. It can keep your joints flexible and reduce stress on them. If the pain does not get better, tell your doctor or nurse. They may talk to you about changing to a different hormonal drug.

Tingling in your fingers or toes

This treatment can affect the nervous system. You may have pins and needles or feel tingling in your hands and feet. This is usually mild. If you notice these symptoms, contact the hospital on the 24-hour number as soon as possible.

Weight gain

You may gain weight when you are having this treatment. [Eating healthily and being active](#) can help you keep to a healthy weight. Your doctor, nurse or dietitian can give you more advice.

Effects on the heart

This treatment might increase certain risk factors for [heart problems](#). Tell your doctor if you have:

- had any problems with your heart
-

high blood pressure

- high levels of cholesterol – a fatty substance in your blood.

They can monitor or treat these if needed. Talk to your doctor if you are worried about this.

Things you can do to take care of your heart include:

- not smoking
- maintaining a healthy weight
- being physically active.

Tumour flare

If you have symptoms caused by the cancer, they may get worse for a short time when you start taking it. This is called tumour flare. Tell your doctor, nurse or pharmacist if this happens, they can give you treatment to help with this.

High calcium levels in the blood

If you have goserelin to treat breast cancer that has spread to the bones, sometimes it may cause high calcium levels in the blood (hypercalcaemia). This is not common. Signs of this include:

- [feeling sick or being sick \(vomiting\)](#)
- being very thirsty
- being [constipated](#)
- being confused.

If you have any of these symptoms, contact the hospital straight away on the 24-hour number.

Allergic reaction

This treatment might cause an allergic reaction. But this is not common. Signs of a reaction can include:

- a skin rash or itching
- feeling breathless or wheezy
- swelling of your face, mouth or throat.

If you feel unwell or have any of these signs, contact the hospital straight away on the 24-hour number. Do not take any more of this treatment until you have checked with them.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Effects on the liver

This treatment may affect how your liver works. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your liver is working.

Other important information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

Date reviewed

Reviewed: 01 May 2026 | Next review: 01 May 2028

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

